Fig_rates

Claire Morton

2026-06-10

Functions

Figures

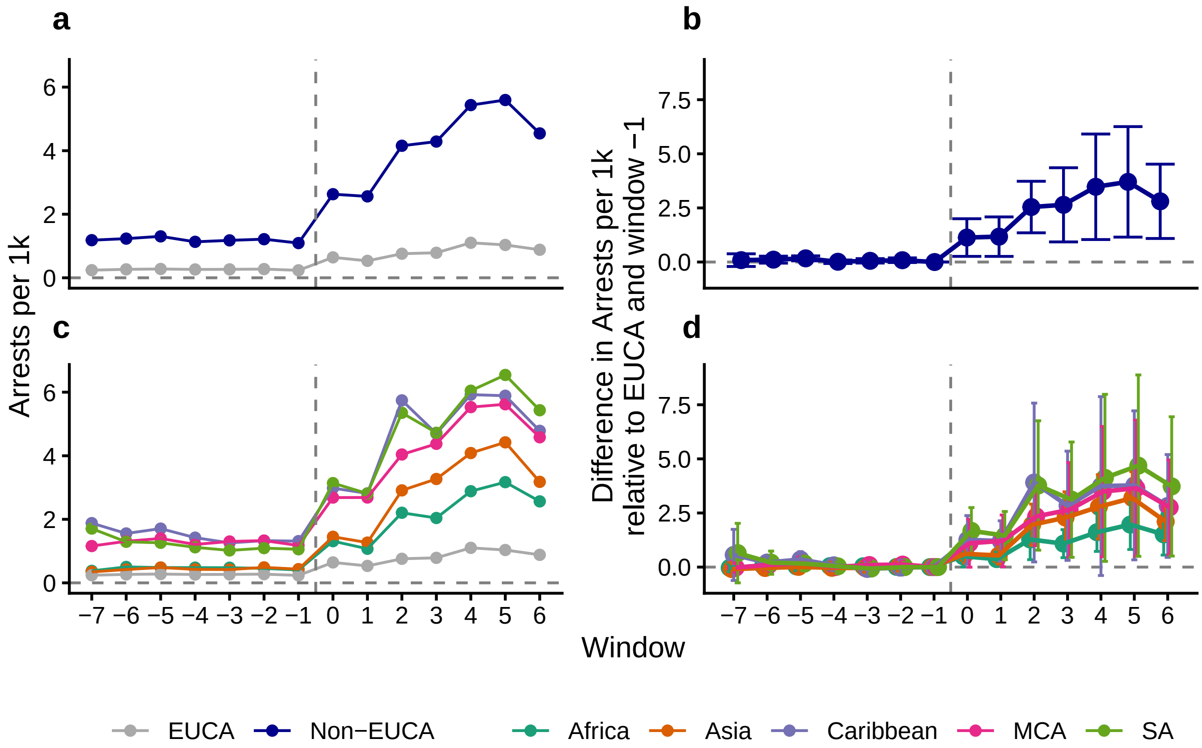
```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## # A tibble: 70 x 6
##    window      est     se   lower upper region
##     <dbl>    <dbl>  <dbl>   <dbl> <dbl> <chr>
##  1     -7 -0.0336  0.0641 -0.214  0.146 Africa
##  2     -6  0.0639  0.0496 -0.0753 0.203 Africa
##  3     -5  0.0332  0.0698 -0.163  0.229 Africa
##  4     -4  0.0455  0.0509 -0.0973 0.188 Africa
##  5     -3  0.0396  0.0398 -0.0720 0.151 Africa
##  6     -2  0.00992 0.0438 -0.113  0.133 Africa
##  7     -1  0       0       0      0     Africa
##  8      0  0.498   0.173   0.0118 0.984 Africa
##  9      1  0.368   0.115   0.0437 0.692 Africa
## 10      2  1.28    0.332   0.344  2.21  Africa
## # i 60 more rows
## Raw coef names:
##  window::-7:region::nonEUCA window::-6:region::nonEUCA window::-5:region::nonEUCA window::-4:region:
## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:r
##  stat = 3.15277, p-value = 0.004485, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"
## $stat
## [1] 3.152774
##
## $p
## [1] 0.004484512
##
## $df1
## [1] 6
##
## $df2
```

```
## [1] 1400
##
## $vcov
## [1] "Custom"
##
## [1] "M bar Results"
## # A tibble: 5 x 5
##        lb    ub method Delta    Mbar
##     <dbl> <dbl> <chr>  <chr>   <dbl>
## 1  0.858   4.32 C-LF   DeltaRM   1
## 2  0.157   5.23 C-LF   DeltaRM   2.2
## 3  0.0846  5.33 C-LF   DeltaRM   2.3
## 4  0.0121  5.40 C-LF   DeltaRM   2.4
## 5 -0.0604  5.50 C-LF   DeltaRM   2.5
## # A tibble: 14 x 6
##    window    est     se   lower  upper region
##     <dbl>  <dbl>  <dbl>   <dbl>  <dbl> <chr>
##  1     -7 0.0870 0.111  -0.206  0.380  nonEUCA
##  2     -6 0.109  0.0608 -0.0517 0.271  nonEUCA
##  3     -5 0.170  0.0436  0.0540 0.285  nonEUCA
##  4     -4 0.0123 0.0309 -0.0697 0.0942 nonEUCA
##  5     -3 0.0553 0.0386 -0.0471 0.158  nonEUCA
##  6     -2 0.0842 0.0406 -0.0234 0.192  nonEUCA
##  7     -1 0      0       0      0      nonEUCA
##  8      0 1.13   0.328   0.261  2.00   nonEUCA
##  9      1 1.17   0.344   0.260  2.08   nonEUCA
## 10      2 2.54   0.449   1.35   3.73   nonEUCA
## 11      3 2.64   0.646   0.930  4.36   nonEUCA
## 12      4 3.48   0.919   1.04   5.91   nonEUCA
## 13      5 3.70   0.963   1.15   6.26   nonEUCA
## 14      6 2.80   0.648   1.09   4.52   nonEUCA

## Warning: Subsetting quosures with `[[` is deprecated as of rlang 0.4.0
## Please use `quo_get_expr()` instead.
## This warning is displayed once every 8 hours.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```




```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.

## # A tibble: 70 x 6
##     window      est      se    lower   upper region
##      <dbl>    <dbl>   <dbl>    <dbl>   <dbl> <chr>
##  1      -7 -0.00518 0.0100  -0.0334  0.0230 Africa
##  2      -6  0.0101  0.00756 -0.0112  0.0314 Africa
##  3      -5  0.00543 0.0107  -0.0248  0.0356 Africa
##  4      -4  0.00724 0.00773 -0.0145  0.0290 Africa
##  5      -3  0.00636 0.00611 -0.0109  0.0236 Africa
##  6      -2  0.00180 0.00685 -0.0175  0.0211 Africa
##  7      -1  0        0        0       0      Africa
##  8       0  0.0801  0.0259   0.00709 0.153  Africa
##  9       1  0.0592  0.0180   0.00835 0.110  Africa
## 10       2  0.202   0.0496   0.0620  0.342  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##     window    est     se    lower  upper region
##      <dbl>  <dbl>  <dbl>    <dbl>  <dbl> <chr>
##  1      -7 0.0286 0.0349 -0.0624  0.120  nonEUCA
##  2      -6 0.0402 0.0174 -0.00510 0.0855 nonEUCA
##  3      -5 0.0613 0.0119  0.0302  0.0925 nonEUCA
```

```
##  4      -4 0.00857 0.00723 -0.0103   0.0274 nonEUCA
##  5      -3 0.0229  0.0117  -0.00746 0.0533 nonEUCA
##  6      -2 0.0335  0.0124   0.00123 0.0657 nonEUCA
##  7      -1 0       0        0        0      nonEUCA
##  8       0 0.431   0.131    0.0886   0.773  nonEUCA
##  9       1 0.425   0.135    0.0748   0.776  nonEUCA
## 10       2 0.901   0.184    0.422    1.38   nonEUCA
## 11       3 0.939   0.250    0.287    1.59   nonEUCA
## 12       4 1.26    0.355    0.333    2.18   nonEUCA
## 13       5 1.32    0.378    0.335    2.30   nonEUCA
## 14       6 1.01    0.259    0.332    1.68   nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
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```
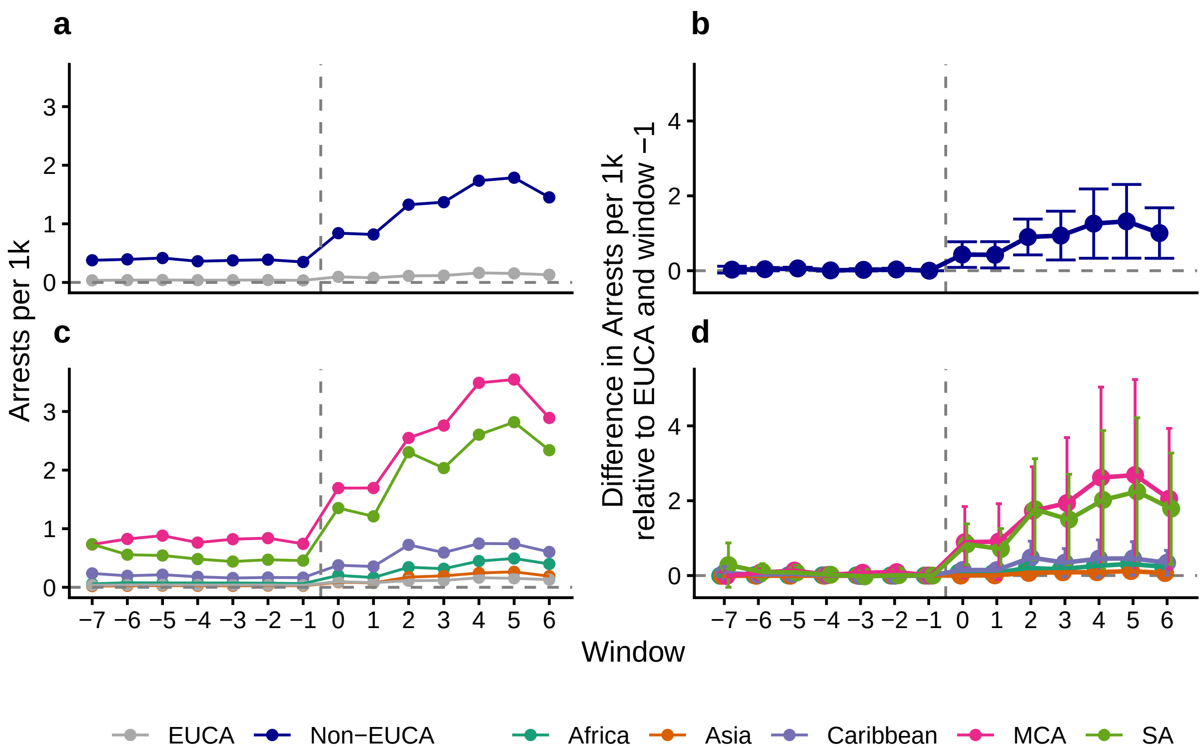
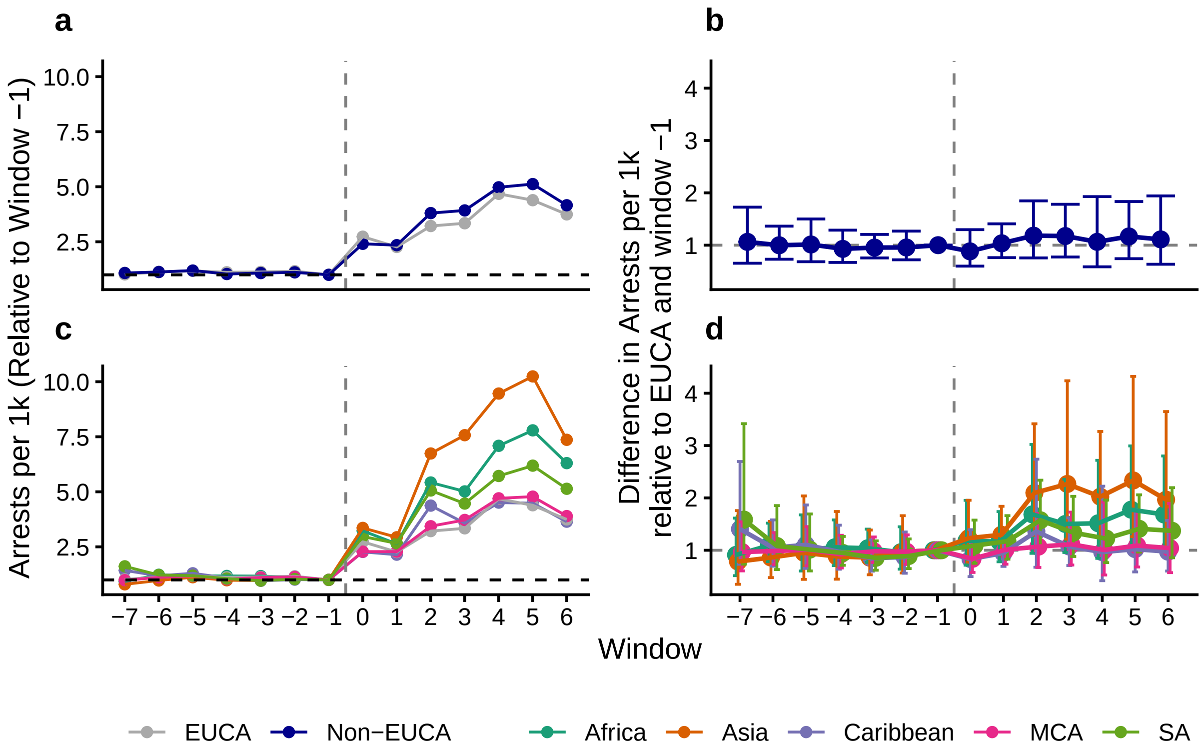




```
## # A tibble: 70 x 6
##    window   est    se lower upper region
##     <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
##  1     -7 0.911 0.204 0.514  1.61 Africa
##  2     -6 1.09  0.118 0.784  1.52 Africa
##  3     -5 1.01  0.183 0.603  1.68 Africa
##  4     -4 1.06  0.144 0.707  1.58 Africa
##  5     -3 1.04  0.108 0.768  1.40 Africa
##  6     -2 0.962 0.146 0.639  1.45 Africa
##  7     -1 1     0     1      1    Africa
##  8      0 1.18  0.180 0.715  1.95 Africa
##  9      1 1.16  0.143 0.780  1.74 Africa
## 10      2 1.69  0.208 0.941  3.02 Africa
## # i 60 more rows
```

```
## Raw coef names:
##  window::-7:region::nonEUCA window::-6:region::nonEUCA window::-5:region::nonEUCA window::-4:region:
## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:re
##  stat = 0.208309, p-value = 0.974287, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"
## $stat
## [1] 0.2083088
##
## $p
## [1] 0.9742866
##
## $df1
## [1] 6
##
## $df2
## [1] 1400
##
## $vcov
## [1] "Custom"

## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate

## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate

## [1] "M bar Results"
## # A tibble: 5 x 5
##      lb    ub method Delta    Mbar
##   <dbl> <dbl> <chr>  <chr>   <dbl>
## 1 -1.30  1.48 C-LF   DeltaRM   1
## 2 -2.76  2.97 C-LF   DeltaRM   2.2
## 3 -2.88  3.09 C-LF   DeltaRM   2.3
## 4 -3.01  3.14 C-LF   DeltaRM   2.4
## 5 -3.13  3.14 C-LF   DeltaRM   2.5
## # A tibble: 14 x 6
##    window   est     se lower upper region
##     <dbl> <dbl>  <dbl> <dbl> <dbl> <chr>
##  1     -7 1.06  0.180  0.655  1.73 nonEUCA
##  2     -6 0.999 0.116  0.731  1.36 nonEUCA
##  3     -5 1.01  0.146  0.684  1.50 nonEUCA
##  4     -4 0.929 0.121  0.670  1.29 nonEUCA
##  5     -3 0.955 0.0864 0.756  1.21 nonEUCA
##  6     -2 0.956 0.105  0.720  1.27 nonEUCA
##  7     -1 1     0      1      1    nonEUCA
##  8      0 0.882 0.143  0.600  1.30 nonEUCA
##  9      1 1.04  0.114  0.762  1.41 nonEUCA
## 10      2 1.18  0.165  0.757  1.85 nonEUCA
## 11      3 1.17  0.154  0.774  1.78 nonEUCA
## 12      4 1.06  0.220  0.587  1.93 nonEUCA
## 13      5 1.17  0.168  0.742  1.83 nonEUCA
## 14      6 1.11  0.207  0.635  1.94 nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



Test for significance

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## [1] "Pre-treatment rates"

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window and region.
## i Output is grouped by window.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## # A tibble: 4 x 5
## # Groups:   window [2]
##   window region  total_arrests      pop  prop
##    <dbl> <chr>           <dbl>    <dbl> <dbl>
## 1     -1 EUCA              208   883000 0.236
## 2     -1 nonEUCA         14015 12829000 1.09
## 3      0 EUCA              724   883000 0.820
## 4      0 nonEUCA         53531. 12829000 4.17
```

```
## [1] 3.544246e-05
```

```
## [1] 2.49684
```

```
## [1] 1.31343
```

```
## [1] 3.68025
```

Test for significance

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
```

```
## [1] "Pre-treatment rates"
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window and region.
## i Output is grouped by window.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 4 x 5
## # Groups:   window [2]
##   window region  total_arrests      pop  prop
##    <dbl> <chr>           <dbl>    <dbl> <dbl>
## 1     -1 EUCA              208   883000 0.236
## 2     -1 nonEUCA         14015 12829000 1.09
## 3      0 EUCA              724   883000 0.820
## 4      0 nonEUCA        53531. 12829000 4.17
```

```
## [1] 0.6477476
```

```
## [1] 1.083454
```

```
## [1] 0.7682085
```

```
## [1] 1.528065
```

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
```

```
## # A tibble: 70 x 6
##    window     est     se   lower  upper region
##     <dbl>   <dbl>  <dbl>   <dbl>  <dbl> <chr>
```

```
##  1     -7  0.0399 0.0494 -0.0971 0.177  Africa
##  2     -6  0.0942 0.0433 -0.0258 0.214  Africa
##  3     -5  0.0918 0.0543 -0.0587 0.242  Africa
##  4     -4  0.0479 0.0471 -0.0828 0.179  Africa
##  5     -3  0.0111 0.0304 -0.0732 0.0954 Africa
##  6     -2 -0.0138 0.0270 -0.0888 0.0612 Africa
##  7     -1  0      0       0      0      Africa
##  8      0  0.359  0.155  -0.0721 0.789  Africa
##  9      1  0.317  0.102   0.0337 0.600  Africa
## 10      2  0.962  0.259   0.244  1.68   Africa
## # i 60 more rows
## # A tibble: 14 x 6
##    window    est     se    lower  upper region
##     <dbl>  <dbl>  <dbl>    <dbl>  <dbl> <chr>
##  1     -7 0.100  0.0402 -0.00838 0.209  nonEUCA
##  2     -6 0.121  0.0330  0.0313  0.210  nonEUCA
##  3     -5 0.124  0.0231  0.0615  0.187  nonEUCA
##  4     -4 0.0426 0.0103  0.0147  0.0705 nonEUCA
##  5     -3 0.0280 0.0141 -0.0102  0.0662 nonEUCA
##  6     -2 0.0260 0.0155 -0.0158  0.0679 nonEUCA
##  7     -1 0      0       0       0      nonEUCA
##  8      0 0.620  0.151   0.214   1.03   nonEUCA
##  9      1 0.558  0.133   0.198   0.917  nonEUCA
## 10      2 1.52   0.183   1.03    2.01   nonEUCA
## 11      3 1.59   0.386   0.546   2.63   nonEUCA
## 12      4 2.09   0.568   0.560   3.63   nonEUCA
## 13      5 2.29   0.607   0.649   3.93   nonEUCA
## 14      6 1.63   0.378   0.603   2.65   nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
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```

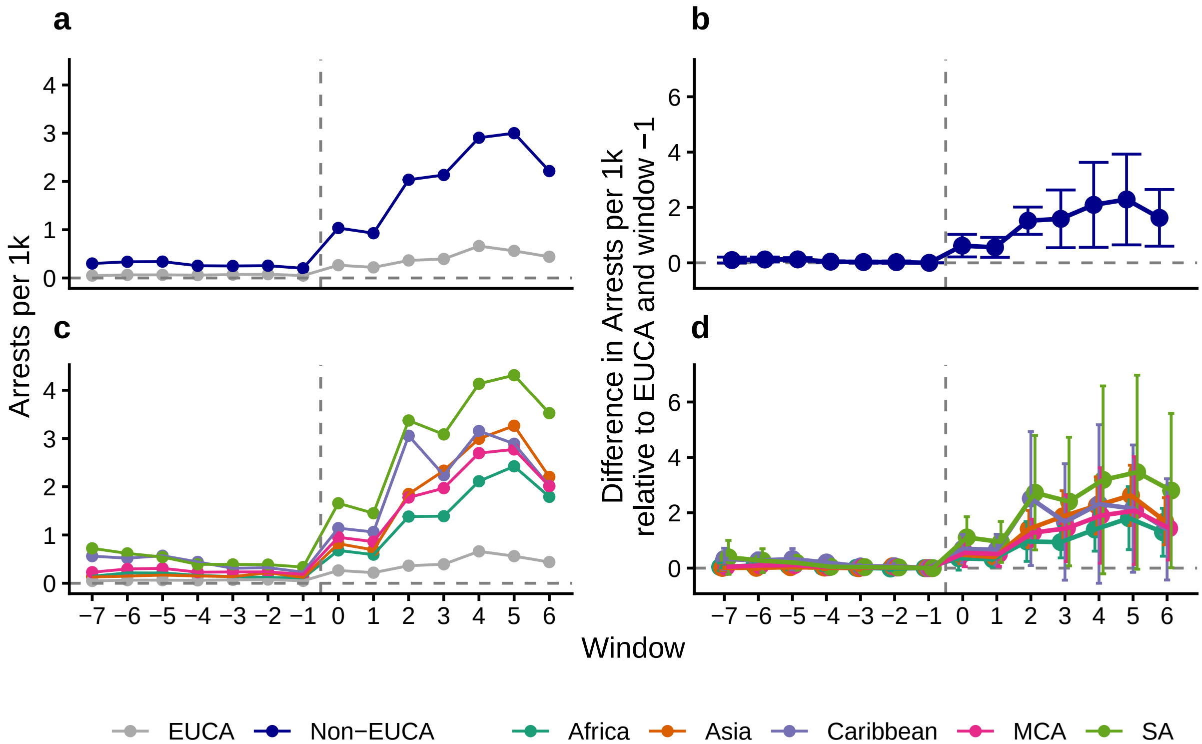


```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.

## # A tibble: 70 x 6
##    window      est      se     lower   upper region
##     <dbl>    <dbl>   <dbl>     <dbl>   <dbl> <chr>
##  1     -7  0.00618 0.00750 -0.0147   0.0271  Africa
##  2     -6  0.0147  0.00639 -0.00305  0.0325  Africa
##  3     -5  0.0144  0.00819 -0.00846  0.0372  Africa
##  4     -4  0.00751 0.00720 -0.0125   0.0276  Africa
##  5     -3  0.00186 0.00468 -0.0112   0.0149  Africa
##  6     -2 -0.00195 0.00424 -0.0138   0.00986 Africa
##  7     -1  0       0        0        0       Africa
##  8      0  0.0572  0.0236  -0.00859  0.123   Africa
##  9      1  0.0503  0.0155   0.00727  0.0934  Africa
## 10      2  0.152   0.0380   0.0458   0.257   Africa
## # i 60 more rows
## # A tibble: 14 x 6
##    window    est      se      lower  upper region
##     <dbl>  <dbl>   <dbl>      <dbl>  <dbl> <chr>
##  1     -7 0.0318 0.0117  0.00000139 0.0636 nonEUCA
##  2     -6 0.0410 0.00956 0.0150     0.0670 nonEUCA
##  3     -5 0.0421 0.00653 0.0244     0.0599 nonEUCA
```

```
## 4      -4 0.0153 0.00332 0.00632    0.0244 nonEUCA
## 5      -3 0.0124 0.00342 0.00314    0.0217 nonEUCA
## 6      -2 0.0132 0.00373 0.00302    0.0233 nonEUCA
## 7      -1 0      0       0          0      nonEUCA
## 8       0 0.235  0.0596  0.0729     0.397  nonEUCA
## 9       1 0.207  0.0535  0.0617     0.353  nonEUCA
## 10      2 0.540  0.0760  0.333      0.746  nonEUCA
## 11      3 0.566  0.147   0.167      0.966  nonEUCA
## 12      4 0.774  0.215   0.188      1.36   nonEUCA
## 13      5 0.819  0.229   0.196      1.44   nonEUCA
## 14      6 0.586  0.145   0.192      0.980  nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
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## Adding another scale for colour, which will replace the existing scale.
```
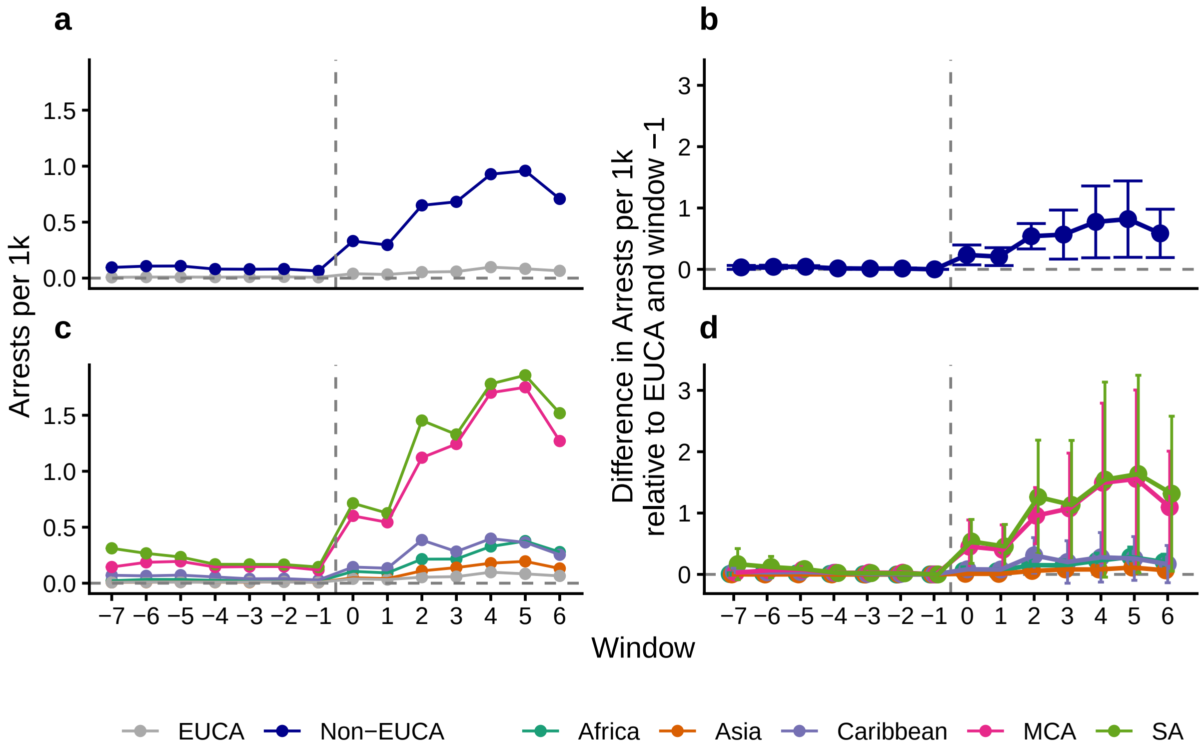
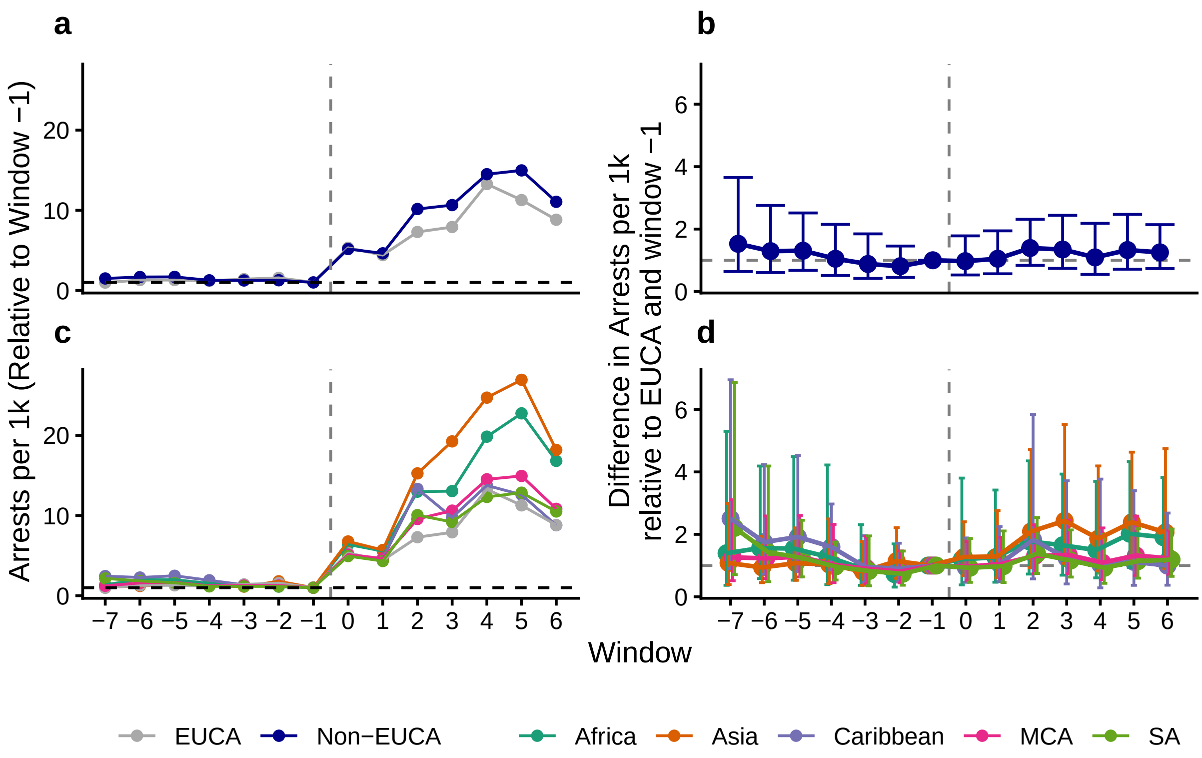



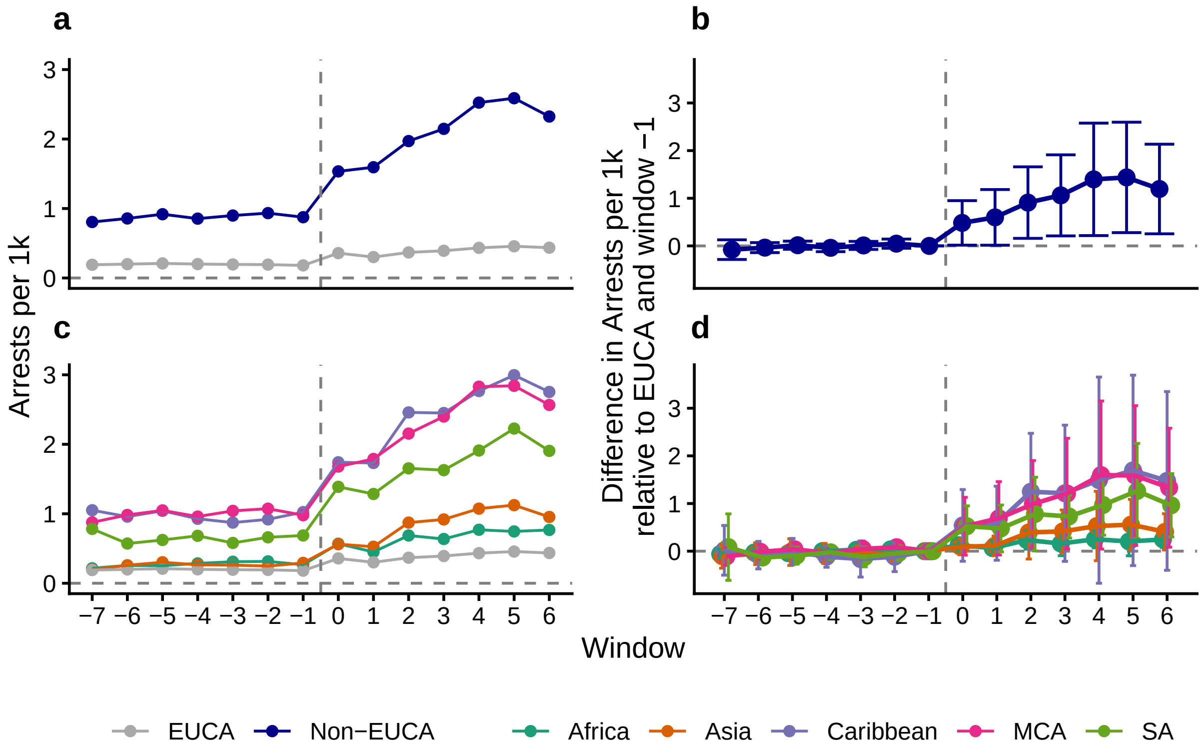

```
## # A tibble: 70 x 6
##    window   est    se lower upper region
##     <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
##  1     -7 1.40  0.513 0.367  5.30 Africa
##  2     -6 1.56  0.379 0.582  4.19 Africa
##  3     -5 1.54  0.410 0.531  4.49 Africa
##  4     -4 1.28  0.457 0.390  4.22 Africa
##  5     -3 0.919 0.354 0.366  2.31 Africa
##  6     -2 0.725 0.326 0.310  1.69 Africa
##  7     -1 1     0     1      1    Africa
##  8      0 1.20  0.443 0.379  3.80 Africa
##  9      1 1.26  0.382 0.466  3.42 Africa
## 10      2 1.78  0.344 0.727  4.35 Africa
## # i 60 more rows
```

```
## # A tibble: 14 x 6
##    window   est    se lower upper region
##     <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
##  1     -7 1.53  0.329 0.640  3.65 nonEUCA
##  2     -6 1.29  0.286 0.606  2.76 nonEUCA
##  3     -5 1.31  0.248 0.678  2.52 nonEUCA
##  4     -4 1.05  0.272 0.511  2.15 nonEUCA
##  5     -3 0.881 0.280 0.421  1.85 nonEUCA
##  6     -2 0.811 0.222 0.451  1.46 nonEUCA
##  7     -1 1     0     1      1    nonEUCA
##  8      0 0.972 0.229 0.530  1.78 nonEUCA
##  9      1 1.05  0.233 0.568  1.94 nonEUCA
## 10      2 1.39  0.192 0.838  2.31 nonEUCA
## 11      3 1.35  0.225 0.743  2.44 nonEUCA
## 12      4 1.09  0.262 0.547  2.18 nonEUCA
## 13      5 1.33  0.235 0.714  2.47 nonEUCA
## 14      6 1.25  0.202 0.734  2.14 nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



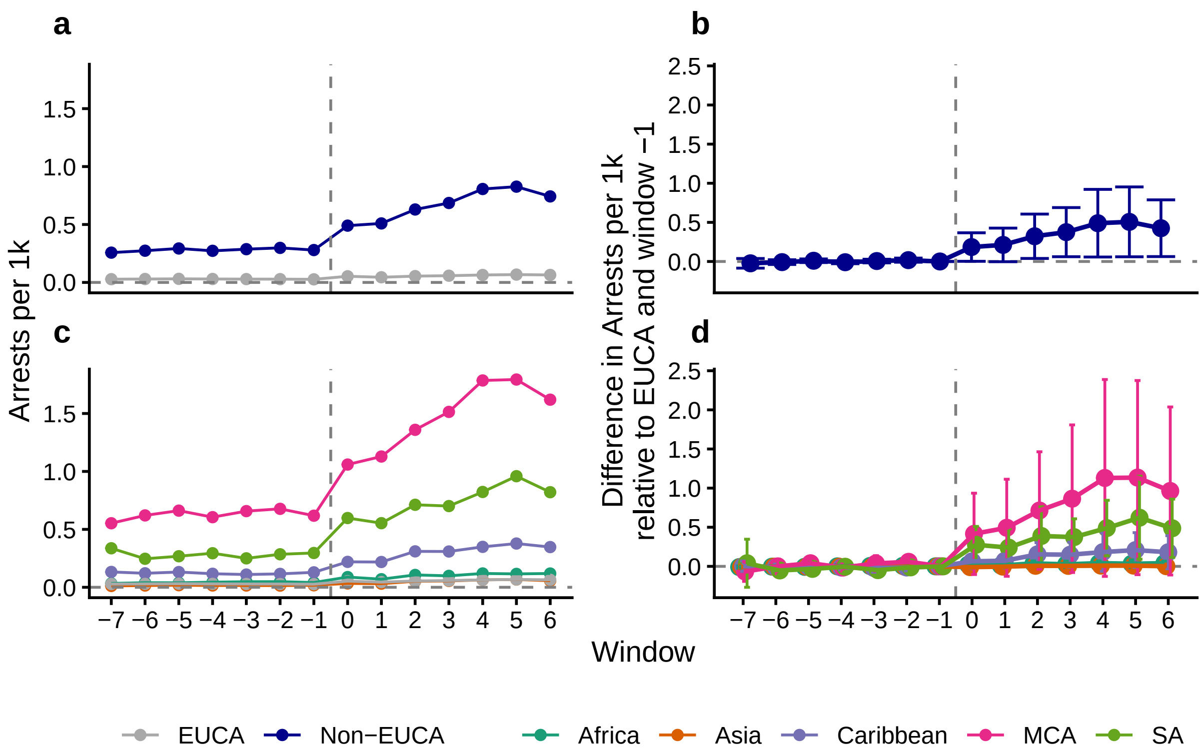

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## # A tibble: 70 x 6
##    window       est     se    lower  upper region
##     <dbl>     <dbl>  <dbl>    <dbl>  <dbl> <chr>
##  1     -7 -0.0599   0.0359 -0.161   0.0415 Africa
##  2     -6 -0.0278   0.0310 -0.115   0.0599 Africa
##  3     -5 -0.0428   0.0388 -0.152   0.0668 Africa
##  4     -4  0.000118 0.0402 -0.114   0.114  Africa
##  5     -3  0.0300   0.0375 -0.0759  0.136  Africa
##  6     -2  0.0371   0.0483 -0.0993  0.174  Africa
##  7     -1  0        0       0       0      Africa
##  8      0  0.126    0.0463 -0.00474 0.257  Africa
##  9      1  0.0640   0.0539 -0.0881  0.216  Africa
## 10      2  0.234    0.0684  0.0414  0.428  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##    window     est     se   lower  upper region
##     <dbl>   <dbl>  <dbl>   <dbl>  <dbl> <chr>
##  1     -7 -0.0784 0.0775 -0.285  0.128  nonEUCA
##  2     -6 -0.0360 0.0394 -0.141  0.0689 nonEUCA
##  3     -5  0.0145 0.0321 -0.0710 0.1000 nonEUCA
##  4     -4 -0.0400 0.0303 -0.121  0.0406 nonEUCA
##  5     -3  0.0103 0.0318 -0.0744 0.0949 nonEUCA
##  6     -2  0.0472 0.0361 -0.0490 0.143  nonEUCA
##  7     -1  0      0       0      0      nonEUCA
##  8      0  0.483  0.176   0.0158 0.950  nonEUCA
##  9      1  0.599  0.220   0.0148 1.18   nonEUCA
## 10      2  0.909  0.282   0.158  1.66   nonEUCA
## 11      3  1.06   0.320   0.210  1.91   nonEUCA
## 12      4  1.40   0.443   0.217  2.58   nonEUCA
## 13      5  1.44   0.436   0.278  2.60   nonEUCA
## 14      6  1.19   0.353   0.254  2.13   nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



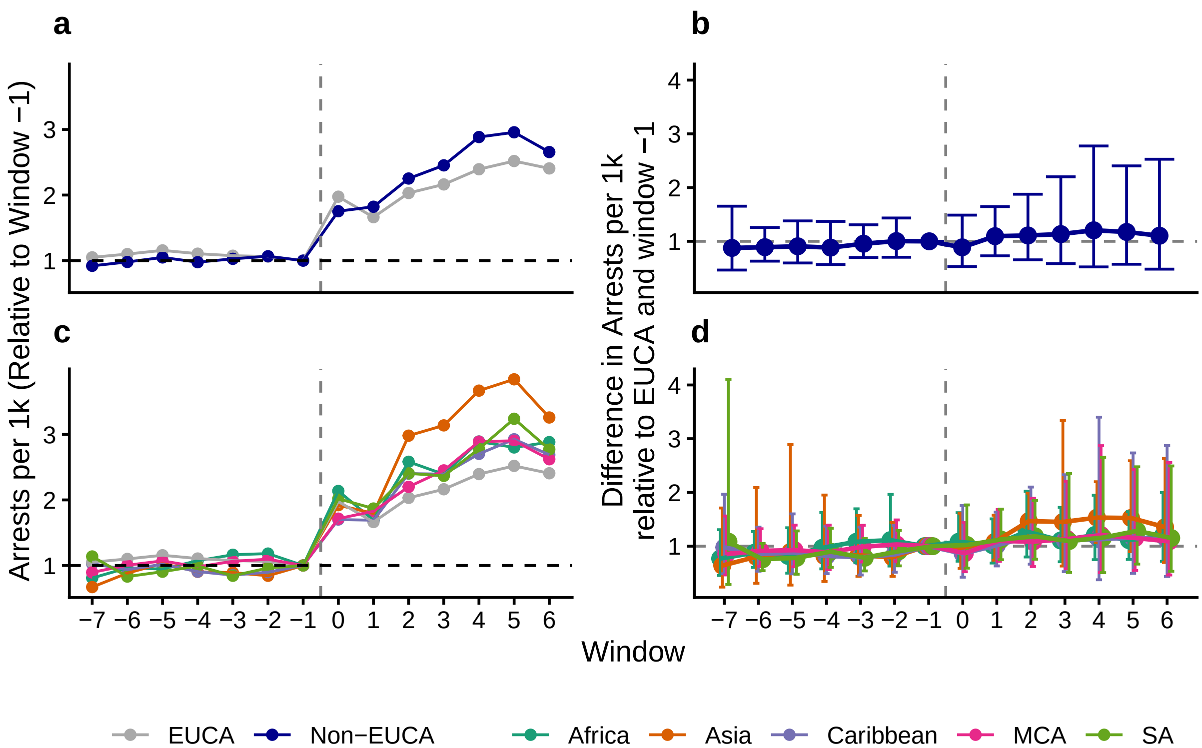

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.

## # A tibble: 70 x 6
##    window       est      se     lower    upper region
##     <dbl>     <dbl>   <dbl>     <dbl>    <dbl> <chr>
##  1     -7 -0.00924  0.00541 -0.0247   0.00618 Africa
##  2     -6 -0.00419  0.00478 -0.0178   0.00942 Africa
##  3     -5 -0.00646  0.00593 -0.0234   0.0104  Africa
##  4     -4  0.000150 0.00615 -0.0174   0.0177  Africa
##  5     -3  0.00475  0.00582 -0.0118   0.0213  Africa
##  6     -2  0.00584  0.00756 -0.0157   0.0274  Africa
##  7     -1  0        0        0        0       Africa
##  8      0  0.0208   0.00709  0.000566 0.0410  Africa
##  9      1  0.0108   0.00865 -0.0139   0.0354  Africa
## 10      2  0.0377   0.0113   0.00555  0.0698  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##    window      est      se    lower   upper region
##     <dbl>    <dbl>   <dbl>    <dbl>   <dbl> <chr>
##  1     -7 -0.0235  0.0238  -0.0849  0.0379  nonEUCA
##  2     -6 -0.00839 0.0118  -0.0387  0.0219  nonEUCA
##  3     -5  0.00947 0.00900 -0.0137  0.0327  nonEUCA
```

```
## 4      -4 -0.00948 0.00751 -0.0289   0.00990 nonEUCA
## 5      -3  0.00560 0.00903 -0.0177   0.0289  nonEUCA
## 6      -2  0.0170  0.0102  -0.00927 0.0433  nonEUCA
## 7      -1  0        0        0        0        nonEUCA
## 8       0  0.184   0.0707   0.00215 0.367   nonEUCA
## 9       1  0.212   0.0833  -0.00302 0.427   nonEUCA
## 10      2  0.322   0.110    0.0388  0.606   nonEUCA
## 11      3  0.375   0.122    0.0609  0.689   nonEUCA
## 12      4  0.489   0.168    0.0570  0.922   nonEUCA
## 13      5  0.506   0.173    0.0597  0.953   nonEUCA
## 14      6  0.425   0.141    0.0625  0.788   nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



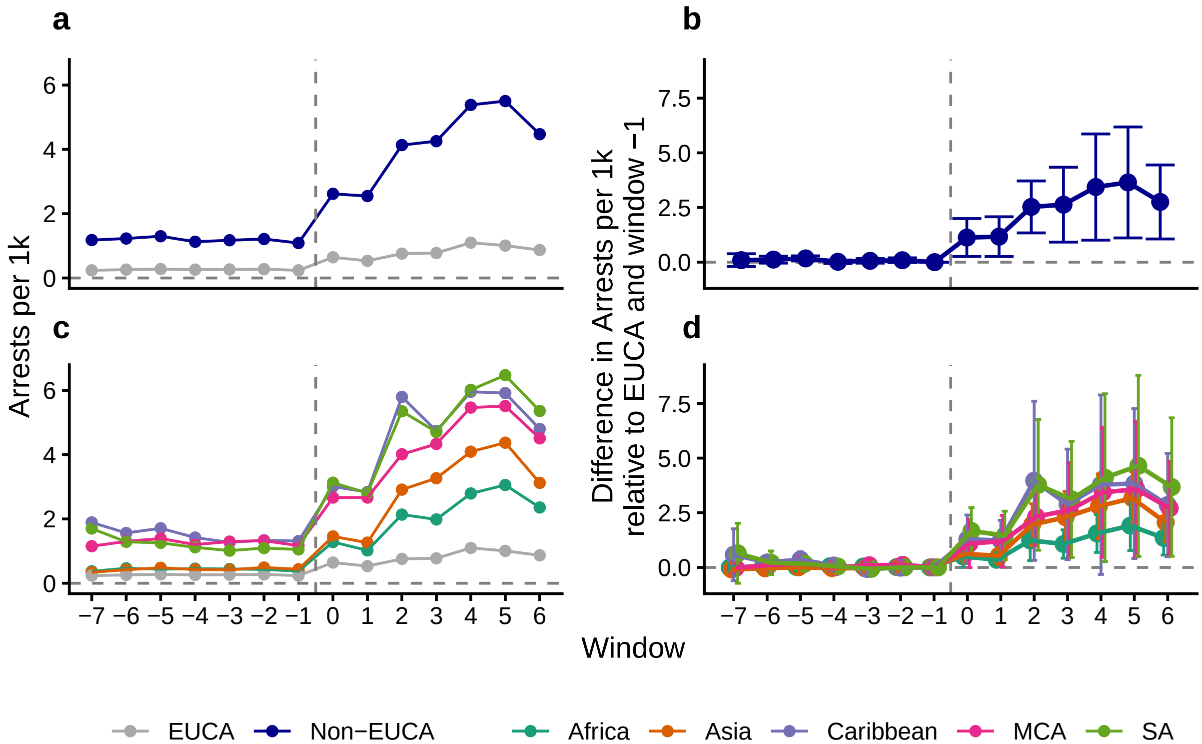

```
## # A tibble: 70 x 6
##    window   est    se lower upper region
##     <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1      -7 0.771 0.190 0.454  1.31 Africa
## 2      -6 0.876 0.134 0.603  1.27 Africa
## 3      -5 0.818 0.179 0.498  1.34 Africa
## 4      -4 0.970 0.187 0.578  1.63 Africa
## 5      -3 1.08  0.162 0.691  1.70 Africa
## 6      -2 1.11  0.205 0.630  1.96 Africa
## 7      -1 1     0     1      1    Africa
## 8       0 1.08  0.146 0.721  1.62 Africa
## 9       1 1.02  0.142 0.686  1.51 Africa
## 10      2 1.27  0.167 0.799  2.02 Africa
## # i 60 more rows
```

```
## # A tibble: 14 x 6
##    window   est    se lower upper region
##     <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
##  1     -7 0.877 0.236 0.465  1.65 nonEUCA
##  2     -6 0.891 0.129 0.631  1.26 nonEUCA
##  3     -5 0.907 0.156 0.597  1.38 nonEUCA
##  4     -4 0.883 0.164 0.568  1.37 nonEUCA
##  5     -3 0.956 0.117 0.699  1.31 nonEUCA
##  6     -2 1.00  0.133 0.703  1.43 nonEUCA
##  7     -1 1     0     1      1    nonEUCA
##  8      0 0.888 0.192 0.530  1.49 nonEUCA
##  9      1 1.10  0.152 0.730  1.65 nonEUCA
## 10      2 1.11  0.196 0.655  1.88 nonEUCA
## 11      3 1.13  0.247 0.585  2.20 nonEUCA
## 12      4 1.21  0.311 0.524  2.77 nonEUCA
## 13      5 1.17  0.267 0.574  2.40 nonEUCA
## 14      6 1.10  0.309 0.482  2.53 nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```




```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## # A tibble: 70 x 6
##    window      est     se    lower upper region
##     <dbl>    <dbl>  <dbl>    <dbl> <dbl> <chr>
##  1     -7 -0.00607 0.0610 -0.177   0.165 Africa
##  2     -6  0.0658  0.0494 -0.0722  0.204 Africa
##  3     -5  0.0216  0.0672 -0.166   0.209 Africa
##  4     -4  0.0535  0.0505 -0.0878  0.195 Africa
##  5     -3  0.0391  0.0387 -0.0692  0.147 Africa
##  6     -2  0.0177  0.0436 -0.104   0.140 Africa
##  7     -1  0       0       0       0     Africa
##  8      0  0.500   0.175   0.00945 0.991 Africa
##  9      1  0.350   0.116   0.0260  0.673 Africa
## 10      2  1.24    0.333   0.307   2.17  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##    window    est     se   lower  upper region
##     <dbl>  <dbl>  <dbl>   <dbl>  <dbl> <chr>
##  1     -7 0.0907 0.111  -0.202  0.384  nonEUCA
##  2     -6 0.117  0.0604 -0.0426 0.277  nonEUCA
##  3     -5 0.171  0.0436  0.0554 0.287  nonEUCA
##  4     -4 0.0173 0.0302 -0.0627 0.0974 nonEUCA
##  5     -3 0.0587 0.0387 -0.0440 0.161  nonEUCA
##  6     -2 0.0881 0.0410 -0.0207 0.197  nonEUCA
##  7     -1 0      0       0      0      nonEUCA
##  8      0 1.12   0.327   0.256  1.99   nonEUCA
##  9      1 1.17   0.343   0.256  2.08   nonEUCA
## 10      2 2.53   0.448   1.34   3.71   nonEUCA
## 11      3 2.63   0.646   0.917  4.34   nonEUCA
## 12      4 3.44   0.916   1.01   5.86   nonEUCA
## 13      5 3.65   0.957   1.11   6.18   nonEUCA
## 14      6 2.75   0.638   1.06   4.45   nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

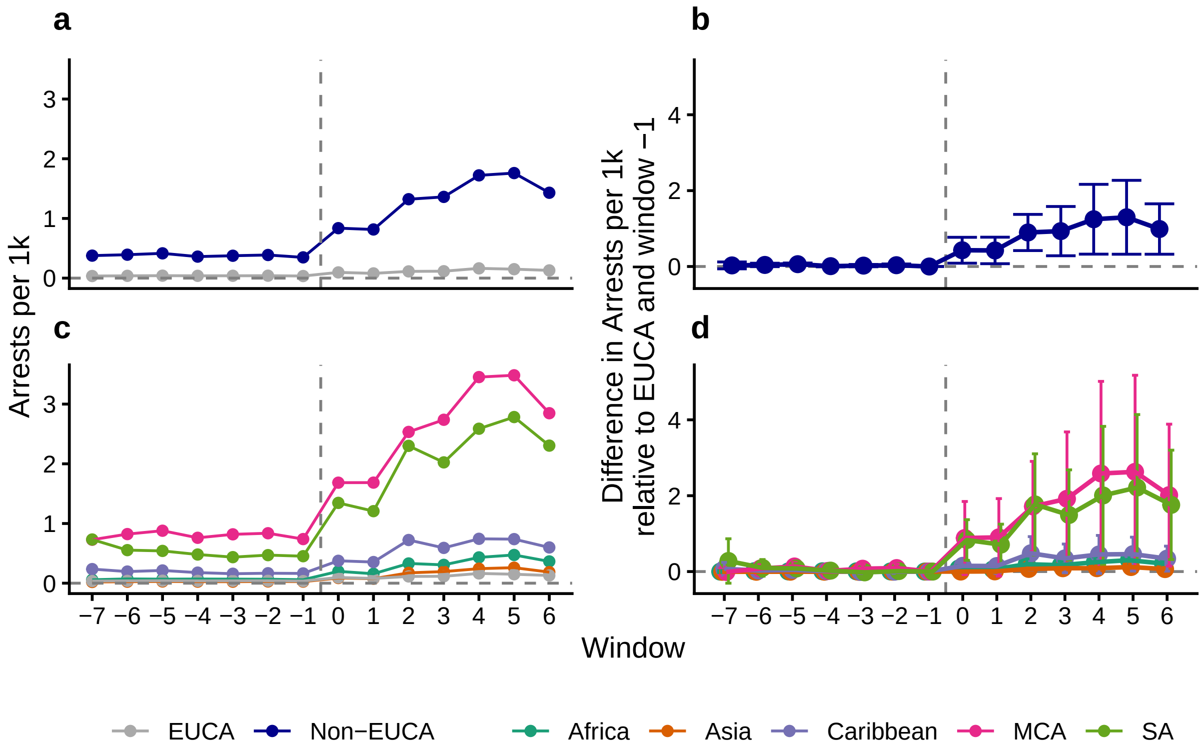



```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.
```

```
## # A tibble: 70 x 6
##    window       est      se     lower  upper region
##     <dbl>     <dbl>   <dbl>     <dbl>  <dbl> <chr>
##  1     -7 -0.000933 0.00942 -0.0276  0.0257 Africa
##  2     -6  0.0103   0.00752 -0.0110  0.0316 Africa
##  3     -5  0.00357  0.0103  -0.0257  0.0328 Africa
##  4     -4  0.00841  0.00766 -0.0133  0.0301 Africa
##  5     -3  0.00621  0.00594 -0.0106  0.0230 Africa
##  6     -2  0.00294  0.00682 -0.0164  0.0222 Africa
##  7     -1  0        0        0       0      Africa
##  8      0  0.0797   0.0262   0.00545 0.154  Africa
##  9      1  0.0558   0.0181   0.00454 0.107  Africa
## 10      2  0.195    0.0498   0.0536  0.336  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##    window    est     se    lower  upper region
##     <dbl>  <dbl>  <dbl>    <dbl>  <dbl> <chr>
##  1     -7 0.0292 0.0350 -0.0617  0.120  nonEUCA
##  2     -6 0.0414 0.0174 -0.00370 0.0866 nonEUCA
##  3     -5 0.0615 0.0120  0.0304  0.0926 nonEUCA
```

```
## 4      -4 0.00944 0.00710 -0.00901 0.0279 nonEUCA
## 5      -3 0.0235  0.0117  -0.00701 0.0539 nonEUCA
## 6      -2 0.0344  0.0126   0.00176 0.0670 nonEUCA
## 7      -1 0       0        0       0      nonEUCA
## 8       0 0.429   0.131    0.0875  0.770  nonEUCA
## 9       1 0.423   0.134    0.0738  0.772  nonEUCA
## 10      2 0.897   0.184    0.420   1.37   nonEUCA
## 11      3 0.933   0.250    0.284   1.58   nonEUCA
## 12      4 1.25    0.354    0.325   2.17   nonEUCA
## 13      5 1.30    0.375    0.323   2.27   nonEUCA
## 14      6 0.988   0.256    0.324   1.65   nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```




```
## # A tibble: 70 x 6
##    window   est    se lower upper region
##     <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
##  1     -7 0.982 0.204 0.559  1.73 Africa
##  2     -6 1.13  0.122 0.803  1.58 Africa
##  3     -5 0.996 0.190 0.589  1.68 Africa
##  4     -4 1.10  0.148 0.728  1.65 Africa
##  5     -3 1.05  0.112 0.773  1.44 Africa
##  6     -2 0.991 0.146 0.662  1.48 Africa
##  7     -1 1     0     1      1    Africa
##  8      0 1.26  0.185 0.751  2.10 Africa
##  9      1 1.21  0.151 0.795  1.84 Africa
## 10      2 1.78  0.218 0.970  3.25 Africa
## # i 60 more rows
```

```
## # A tibble: 14 x 6
##    window   est     se lower upper region
##     <dbl> <dbl>  <dbl> <dbl> <dbl> <chr>
##  1     -7 1.08  0.181  0.663  1.76 nonEUCA
##  2     -6 1.03  0.115  0.755  1.40 nonEUCA
##  3     -5 1.02  0.146  0.688  1.52 nonEUCA
##  4     -4 0.946 0.119  0.686  1.31 nonEUCA
##  5     -3 0.968 0.0854 0.768  1.22 nonEUCA
##  6     -2 0.966 0.104  0.729  1.28 nonEUCA
##  7     -1 1     0      1      1    nonEUCA
##  8      0 0.888 0.144  0.603  1.31 nonEUCA
##  9      1 1.04  0.115  0.766  1.42 nonEUCA
## 10      2 1.19  0.166  0.760  1.87 nonEUCA
## 11      3 1.20  0.156  0.785  1.83 nonEUCA
## 12      4 1.07  0.221  0.589  1.95 nonEUCA
## 13      5 1.19  0.168  0.759  1.88 nonEUCA
## 14      6 1.12  0.208  0.642  1.97 nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```